Technology Considerations

consider the expressions for the first eight carries in CLA:

$c1 = g0 + p0c0$

$c2 = g1 + p1g0 + p1p0c0$

...

$c8 = g7 + p7g6 + p7p6g5 + p7p6p5g4 + p7p6p5p4g3 + p7p6p5p4p3g2 + p7p6p5p4p3p2g1 + p7p6p5p4p3p2p1g0 + p7p6p5p4p3p2p1p0c0$

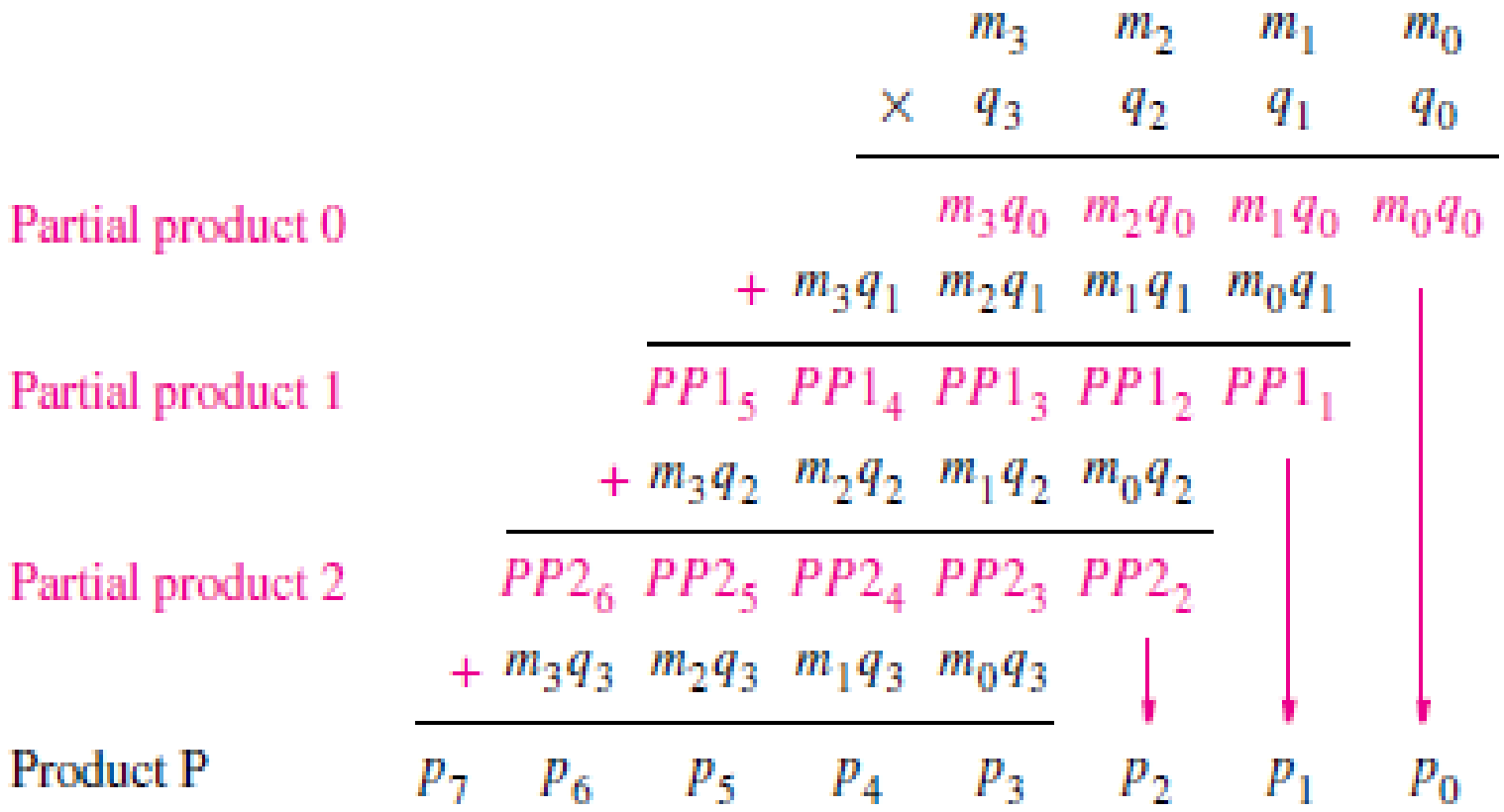
Suppose that the maximum fan-in of the gates is four inputs. Then it is impossible to implement all of these expressions with a two-level AND-OR circuit. The biggest problem is $c8$, where one of the AND gates requires nine inputs; moreover, the OR gate also requires nine inputs. To meet the fan-in constraint, we can rewrite the expression for $c8$ as

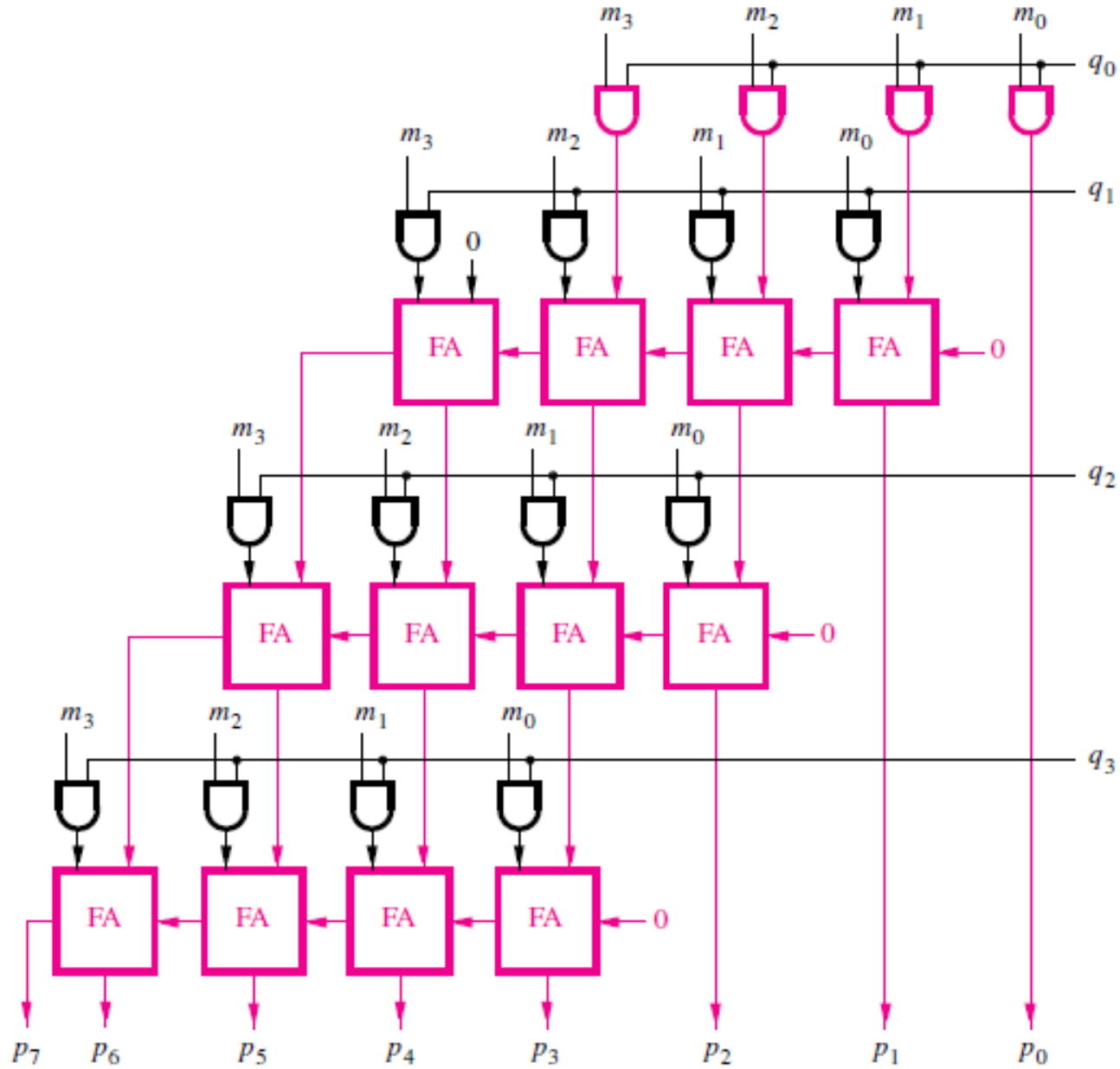
$$c8 = (g7 + p7g6 + p7p6g5 + p7p6p5g4) + [(p7p6p5p4)(g3 + p3g2 + p3p2g1 + p3p2p1g0)] + (p7p6p5p4)(p3p2p1p0)c0$$

To implement this expression we need ten AND gates and three OR gates. The propagation delay in generating $c8$ consists of one gate delay to develop all gi and pi , two gate delays to produce the sum-of-products terms in parentheses, one gate delay to form the product term in square brackets, and one delay for the final ORing of terms. Hence $c8$ is valid after five gate delays, rather than the three gate delays that would be needed without the fan-in constraint.

Multiplication

Two binary numbers can be multiplied using the same method as we use for decimal numbers.




m_3 m_2 m_1 m_0
q_0
q_1
q_2
q_3
0
FA
p_7 p_6 p_5 p_4 p_3 p_2 p_1 p_0

Binary-Coded-Decimal Representation

Table 3.3 Binary-coded decimal digits.

Decimal digit	BCD code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

BCD Addition

X	0 1 1 1	7
+ Y	+ 0 1 0 1	+ 5
Z	1 1 0 0	12
	+ 0 1 1 0	
carry →	1 0 0 1 0	
	S = 2	

X	1 0 0 0	8
+ Y	+ 1 0 0 1	+ 9
Z	1 0 0 0 1	17
	+ 0 1 1 0	
carry →	1 0 1 1 1	
	S = 7	

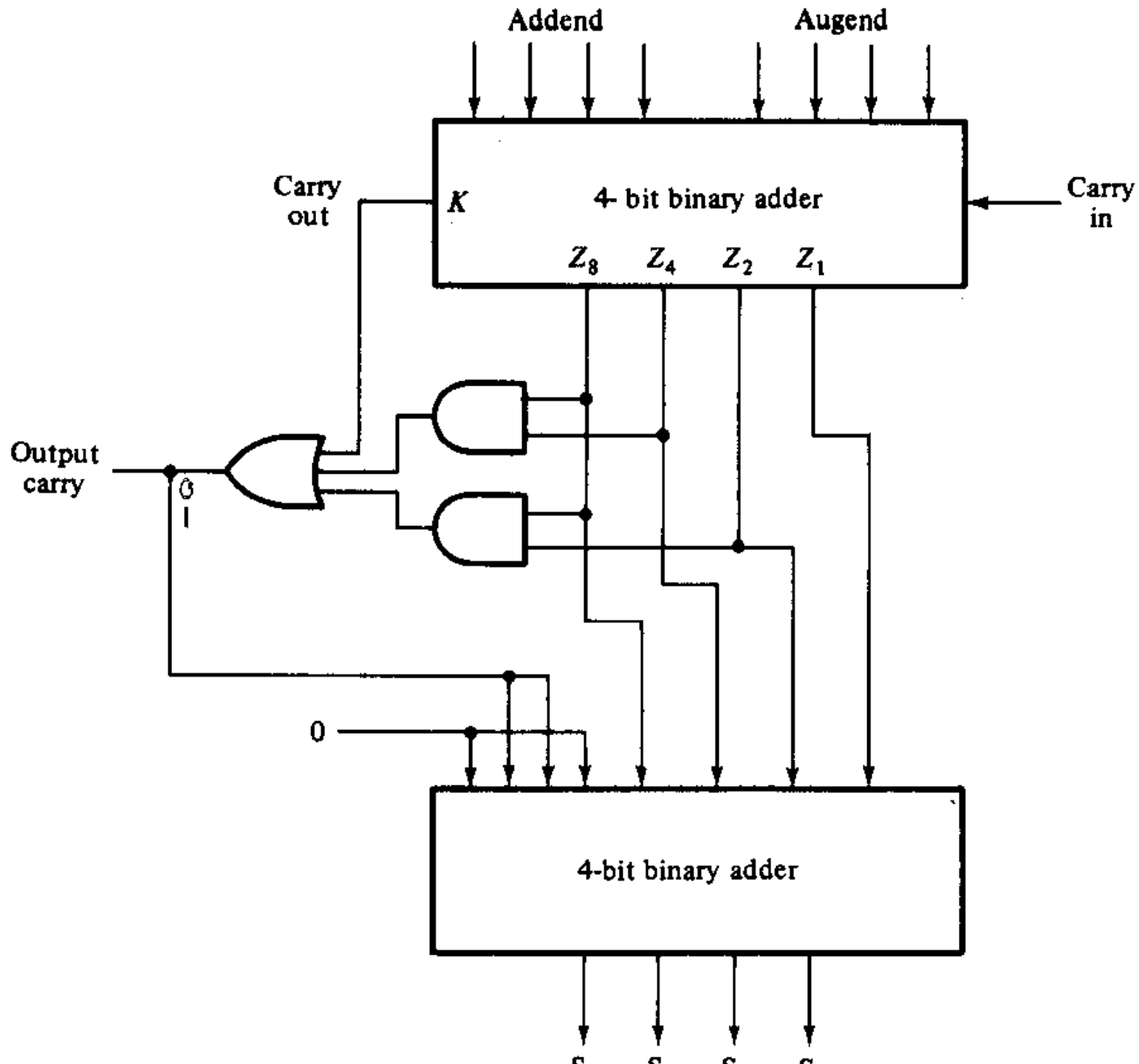
Binary Sum					BCD Sum					Decimal
K	Z_8	Z_4	Z_2	Z_1	C	S_8	S_4	S_2	S_1	
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	2
0	0	0	1	1	0	0	0	1	1	3
0	0	1	0	0	0	0	1	0	0	4
0	0	1	0	1	0	0	1	0	1	5
0	0	1	1	0	0	0	1	1	0	6
0	0	1	1	1	0	0	1	1	1	7
0	1	0	0	0	0	1	0	0	0	8
0	1	0	0	1	0	1	0	0	1	9
0	1	0	1	0	1	0	0	0	0	10
0	1	0	1	1	1	0	0	0	1	11
0	1	1	0	0	1	0	0	1	0	12
0	1	1	0	1	1	0	0	1	1	13
0	1	1	1	0	1	0	1	0	0	14
0	1	1	1	1	1	0	1	0	1	15
1	0	0	0	0	1	0	1	1	0	16
1	0	0	0	1	1	0	1	1	1	17
1	0	0	1	0	1	1	0	0	0	18
1	0	0	1	1	1	1	0	0	1	19

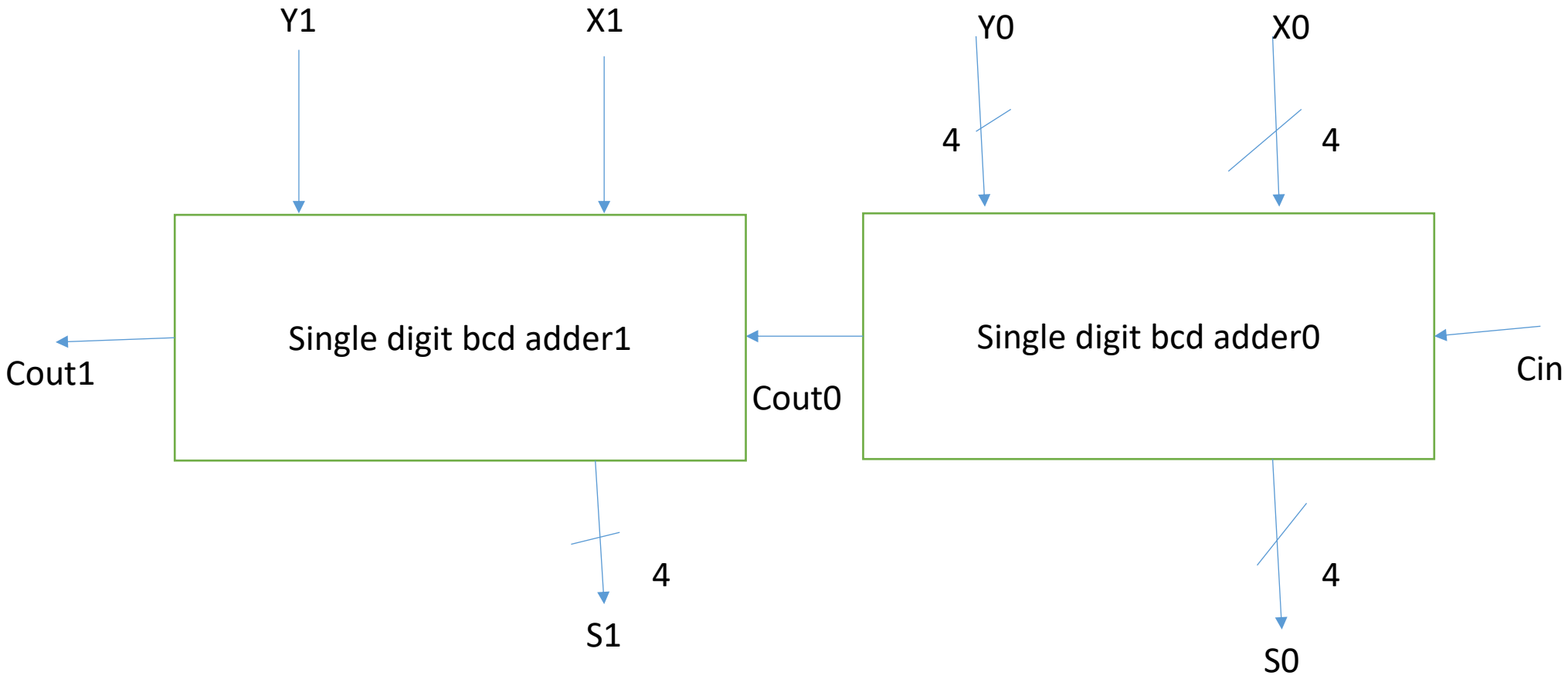
z8, z4 \ z2, z1	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	1	1

Correction should be done when the carry k = 1 or when the expression z8z4 + z8z2 evaluates to 1. Therefore the expression for the correction circuit is

$$C = K + Z_8 Z_4 + Z_8 Z_2$$

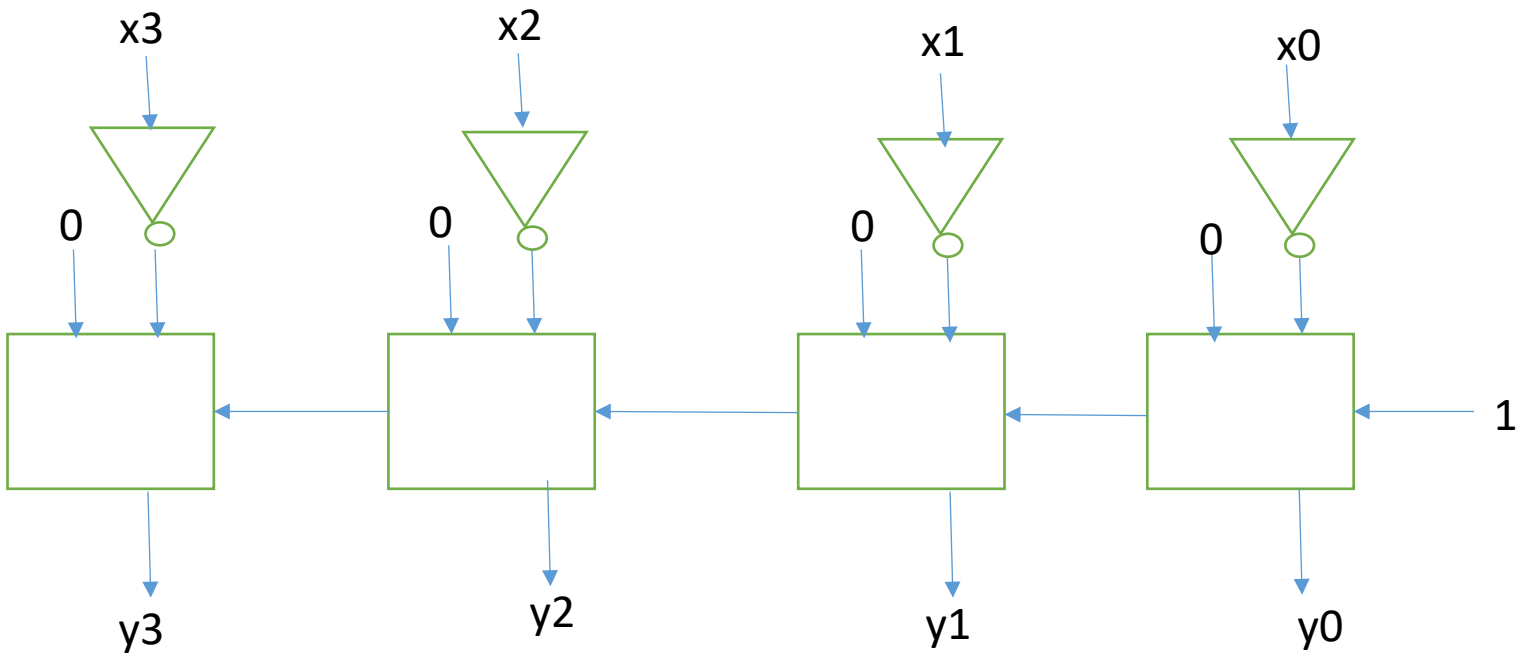
$$C = K + Z_8 Z_4 + Z_8 Z_2$$


Addend
Augend
Carry out
K
4- bit binary adder
Carry in
Z_8
Z_4
Z_2
Z_1
Output carry
0
1
0
4-bit binary adder


Y1
X1
Y0
X0
4
4
Single digit bcd adder1
Single digit bcd adder0
Cout1
Cout0
Cin
4
4
S1
S0

Exercise 1

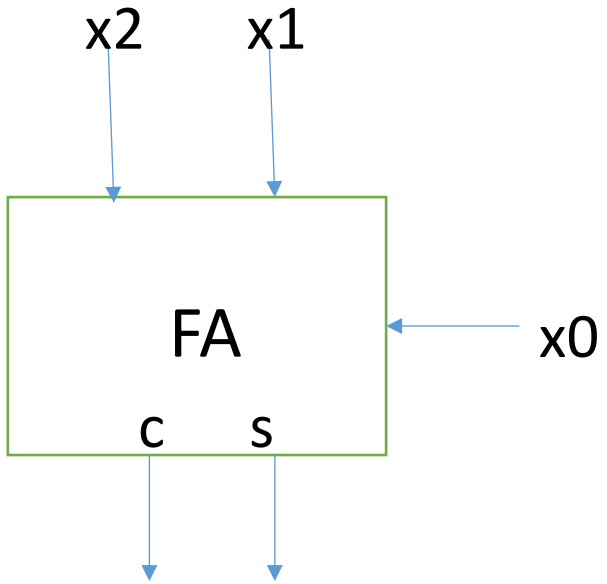
Design a circuit using full adders to generate 2's complement of a 4 bit number X



y3y2y1y0 is the 2's complement of X

Exercise 2

Design a circuit using full adders to find the no. of 1's in a three bit unsigned number



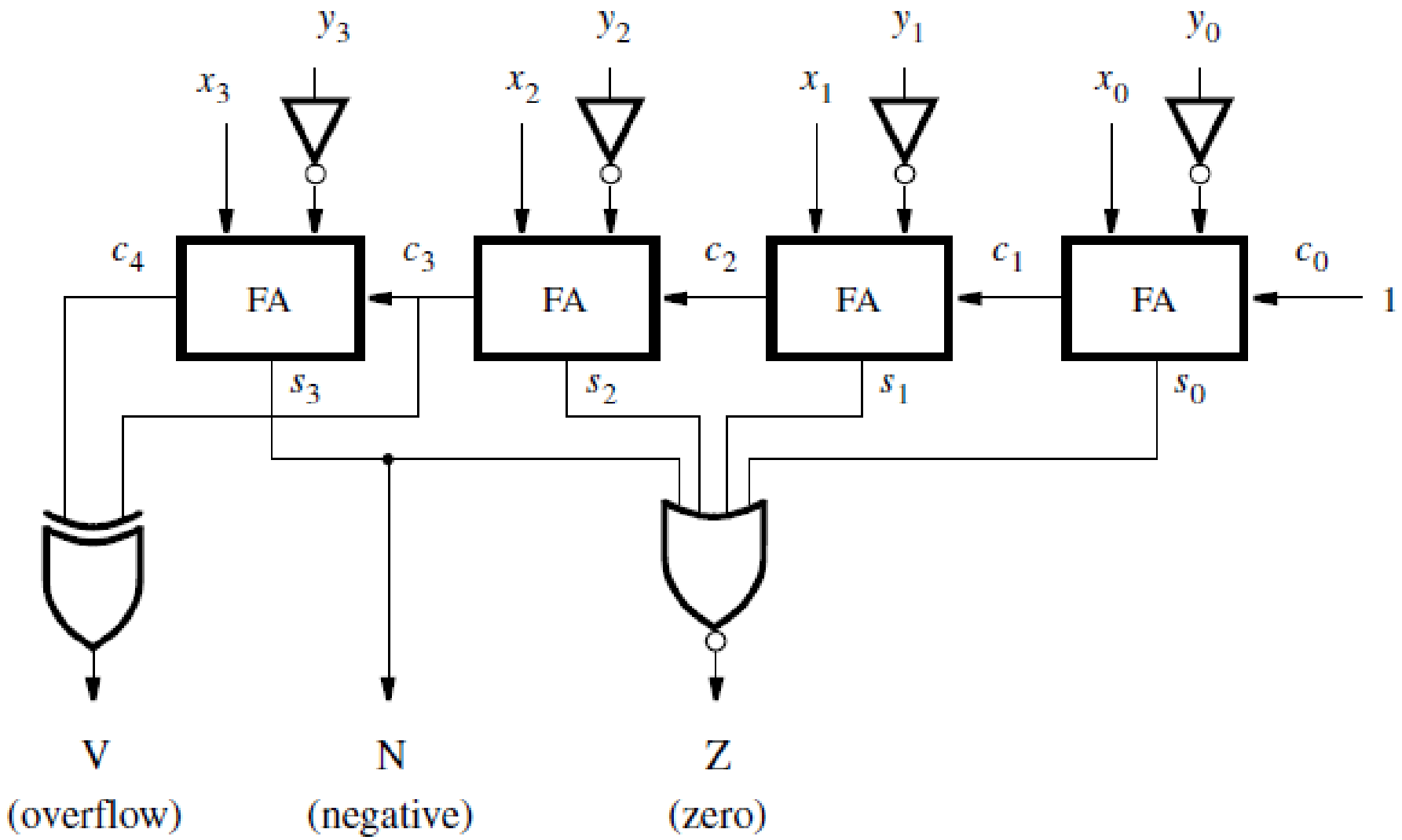
CS is the result in binary.

Example

Problem: In computer computations it is often necessary to compare numbers. Two four-bit signed numbers, $X = x3x2x1x0$ and $Y = y3y2y1y0$, can be compared by using the subtractor circuit. The three outputs denote the following:

- $Z = 1$ if the result is 0; otherwise $Z = 0$
- $N = 1$ if the result is negative; otherwise $N = 0$
- $V = 1$ if arithmetic overflow occurs; otherwise $V = 0$

Show how Z, N, and V can be used to determine the cases $X = Y$, $X < Y$, $X \leq Y$, $X > Y$, and $X \geq Y$.


y_3
y_2
y_1
y_0
x_3
x_2
x_1
x_0
c_4
c_3
c_2
c_1
c_0
FA
FA
FA
FA
1
s_3
s_2
s_1
s_0
V
(overflow)
N
(negative)
Z
(zero)

Consider first the case $X < Y$, where the following possibilities may arise:

- If X and Y have the same sign there will be no overflow, hence $V = 0$. Then for both positive and negative X and Y the difference will be negative ($N = 1$).
- If X is negative and Y is positive, the difference will be negative ($N = 1$) if there is no overflow ($V = 0$); but the result will be positive ($N = 0$) if there is overflow ($V = 1$).

Therefore, if $X < Y$ then $N \oplus V = 1$.

The case $X = Y$ is detected by $Z = 1$. Then, $X \leq Y$ is detected by $Z + (N \oplus V) = 1$.

$X > Y$ if $(Z + (N \oplus V))' = 1$

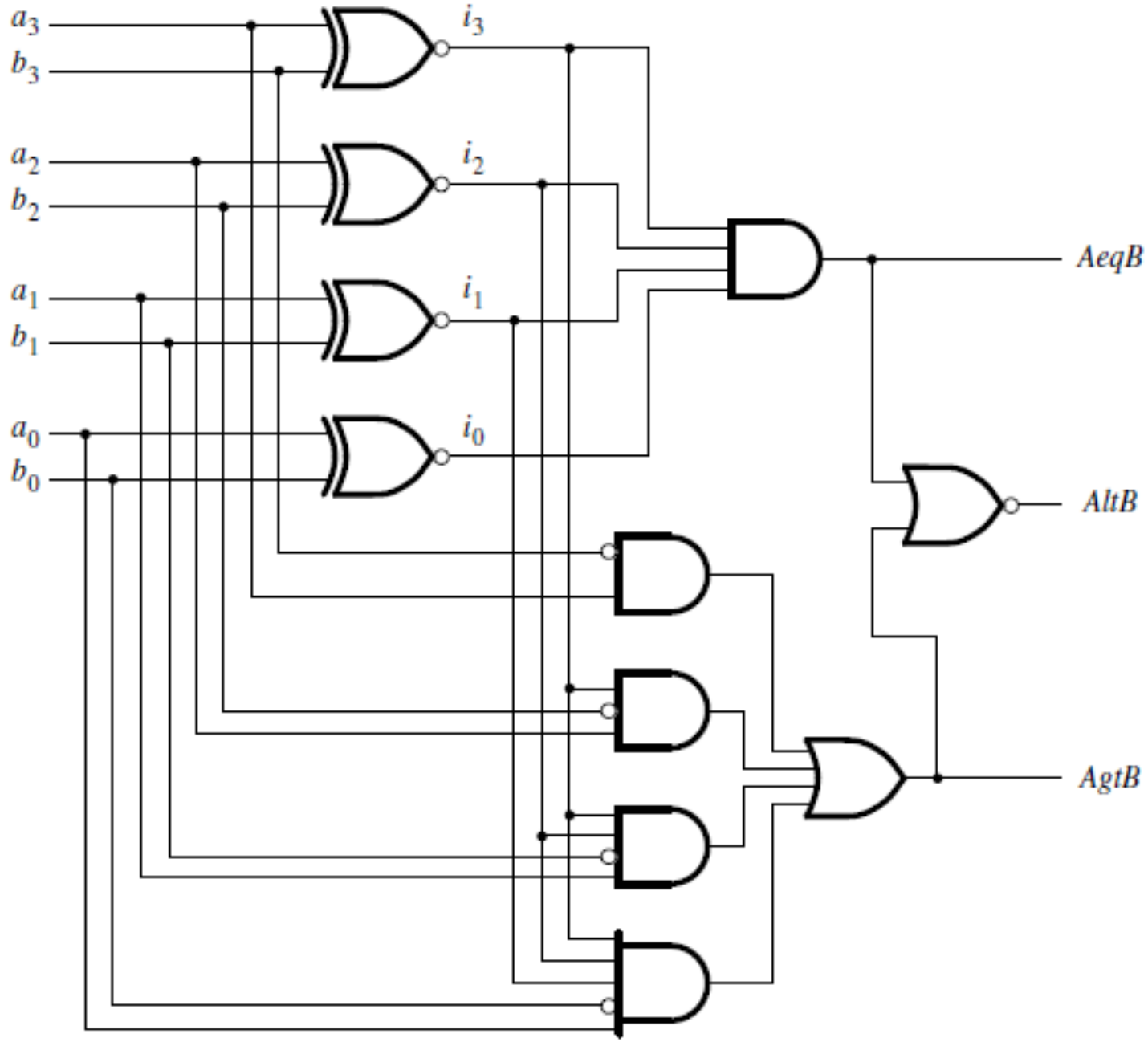
$X \geq Y$ if $(N \oplus V)' = 1$.

Arithmetic Comparison Circuits

Let $A = a3a2a1a0$ and $B = b3b2b1b0$. Define a set of intermediate signals called $i3$, $i2$, $i1$, and $i0$. Each signal, ik , is 1 if the bits of A and B with the same index are equal.
That is, $ik = (ak \oplus bk)'$. The comparator's $AeqB$ output is then given by $AeqB = i3i2i1i0$

$$AgtB = a_3\overline{b}_3 + i_3a_2\overline{b}_2 + i_3i_2a_1\overline{b}_1 + i_3i_2i_1a_0\overline{b}_0$$

$$AltB = \overline{AeqB + AgtB}$$


a_3
b_3
a_2
b_2
a_1
b_1
a_0
b_0
i_3
i_2
i_1
i_0
AeqB
AltB
AgtB